



ECONOMIC RESEARCH
FEDERAL RESERVE BANK OF ST. LOUIS
WORKING PAPER SERIES

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Working Paper Number	2024-027F
Revision Date	October 2025
Citable Link	https://doi.org/10.20955/wp.2024.027
Suggested Citation	Bick, A., Blandin, A., Deming, D.J., 2025; The Rapid Adoption of Generative AI, Federal Reserve Bank of St. Louis Working Paper 2024-027. URL https://doi.org/10.20955/wp.2024.027

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The Rapid Adoption of Generative AI*

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October 27, 2025

Abstract

Generative artificial intelligence (AI) is a potentially important new technology, but its impact on the economy depends on the speed and intensity of adoption. This paper reports results from a series of nationally representative U.S. surveys of generative AI use at work and at home. As of late 2024, 45% of the U.S. population age 18-64 uses generative AI. Among employed respondents, 27% used generative AI for work at least once in the previous week: 10% used it every workday, and 17% on some but not all workdays. Relative to each technology's first mass-market product launch, work adoption of generative AI has been as fast as the personal computer (PC), and overall adoption has been faster than either PCs or the internet. Between 1 and 7% of all work hours are currently assisted by generative AI, and respondents report time savings equivalent to 1.4% of total work hours. Potential productivity gains vary widely by industry, and firm climate and policies play an important role in adoption patterns.

JEL Codes: J24, O33

Keywords: Generative AI, Technology Adoption, Labor Productivity

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Non-employed respondents are not shown the term in brackets. While the CIU asks about computer use “at home”, we use slightly different language to account for respondents who use genAI on mobile devices outside of their home.

The survey asks several follow-up questions of genAI users regarding which products they used, how intensively they used it, and how much time they saved, always distinguishing between work and non-work use.

An important caveat regarding our measurement approach is that it will only capture genAI use that respondents are aware of. For example, while typing prompts into ChatGPT is an obvious case of using genAI, automatic AI-assisted writing suggestions are a more subtle case that respondents may not consider. Because our estimates may not detect some passive or embedded uses of the technology, we interpret our estimates a lower bound on the extent of genAI adoption.

3 How Prevalent is Generative AI Use?

Figure 1a displays the share of August and November 2024 RPS respondents who report using genAI. The samples for the “Overall” and “Outside of Work” bars are all respondents, while the sample for the “For Work” bar is employed respondents. The overall estimates reflect the share who use genAI either for work or outside of work.

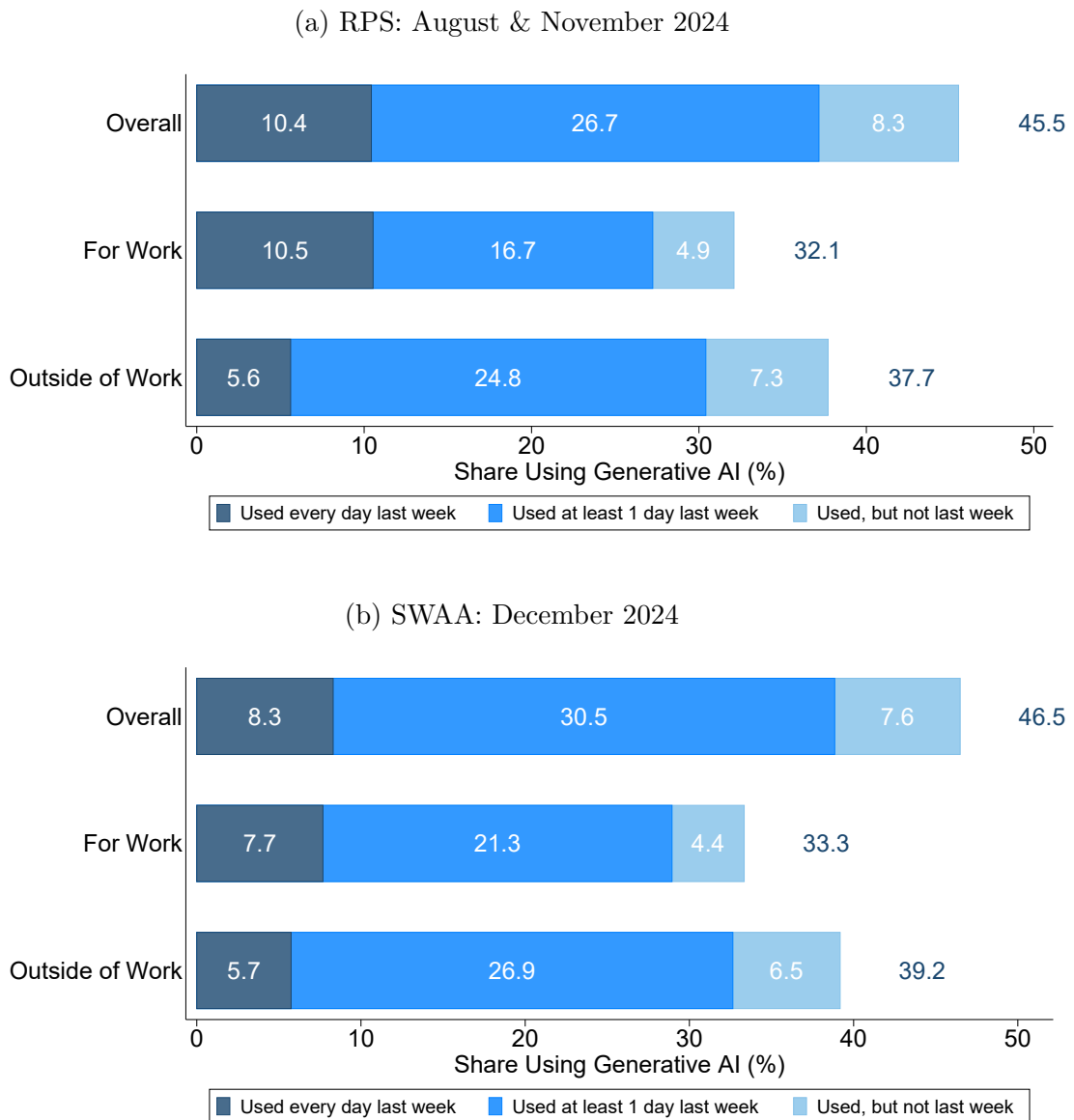
The first bar shows that 45.5% of respondents report using genAI either at work or at home: 10.4% used it every day, 26.7% used it some but not all days, and 8.3% did not use it in the last week. The second bar shows that 32.1% of workers used genAI at work: 10.5% used it every work day and 16.7% on some but not all workdays, bringing total weekly usage to 27.2%. Outside of work, usage is higher overall (37.7%) but less intensive, with only 5.6% of all respondents using it every day last week.

3.1 How Do Our Results Compare to Other Estimates?

The RPS is administered by Qualtrics. To provide a second source of data, in December 2024 we fielded our genAI module through the Survey of Working Arrangements and Attitudes (SWAA) by Barrero et al. (2021).⁵ The SWAA uses a different survey provider (IncQuery) but targets a similar sample intended to be representative of the U.S. population ages 20-64. Figure 1b displays analogous results from the SWAA. 46.5% of SWAA respondents reported using genAI overall, versus 45.5% of RPS respondents. We find similar estimates for work (33.3% SWAA,

⁵See Appendix C for more details on the SWAA, how we construct weights to maximize comparability between the SWAA and RPS, and some validation exercises.

Figure 1: Share of Working Age Adults Using Generative AI



Notes: The figure shows the share of respondents who use genAI for work, outside of work, and overall (either for work or outside of work). Intensity of use is broken down into every day last week (dark blue), at least one day but not every day last week (medium blue), and not last week (light blue). Data source for panel (a) is the August and November 2024 waves of the RPS, ages 18-64. The “For Work” sample is employed individuals ($N = 6935$); the other bars include all respondents ($N = 9726$). Data source for panel (b) is the December 2024 wave of the SWAA, ages 20-64. The “For Work” sample is employed individuals ($N = 3516$); the other bars include all respondents ($N = 4698$).

32.1% RPS) and outside work (39.2% SWAA, 37.7% RPS). Appendix C.3 includes additional comparisons between the SWAA and RPS. Because the two surveys yield similar estimates but the SWAA only includes a subset of the genAI questions in the RPS, the remainder of the main text reports results only from the RPS.